Brackets, squares and square roots

Day 04 • 2026-11-03 • 12:45 PM–2:00 PM

15 questions. Show your method and include units. For multiple choice, enter A, B, C or D. Follow your instructor’s calculator guidance.

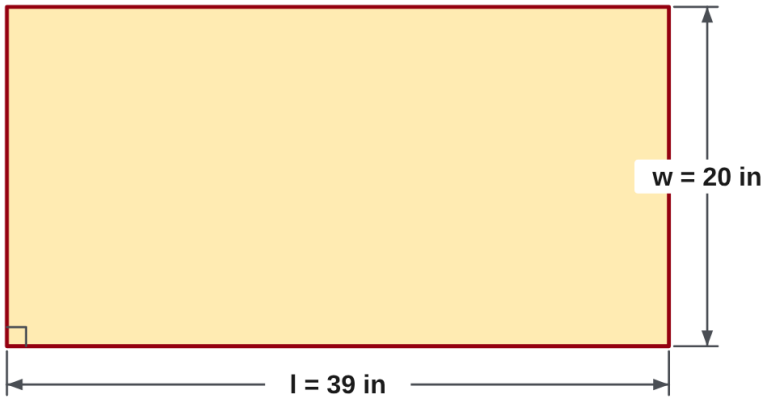
Name _____ **Date** _____

Type into the fields and save a copy, or print and write.

01 READ & REASON

When checking bracket and root steps in Numeracy Hub, when should the worked steps be shown?

A. Only after changing all numbers
B. After making a written attempt
C. Instead of recording any steps
D. Before reading the expression



Numeracy Hub

Choice (A–D)

Working / reason

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

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02 What is the first operation in $(6 + 4)^2$?

A. Multiply 6 by 4

B. Square 4 only

C. Square 6 only

D. Add 6 and 4

Choice (A–D)

Working / reason

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

03 Which statement correctly interprets $\sqrt{81}$?

A. 9, because $9 \times 9 = 81$

B. 162, because a root doubles the number

C. 40.5, because a root halves the number

D. 6, because 8 + 1 is 9 then minus 3

Choice (A–D)

Working / reason

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

04 Which action reverses squaring a positive value?

A. Subtract 2

B. Take the positive square root

C. Divide by 2

D. Double it

Choice (A–D)

Working / reason

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

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05 Which expression squares the entire difference between 15 and 9?

A. $15 - 9^2$

B. $15^2 - 9$

C. $(15 - 9)^2$

D. $15 - 9 \times 2$

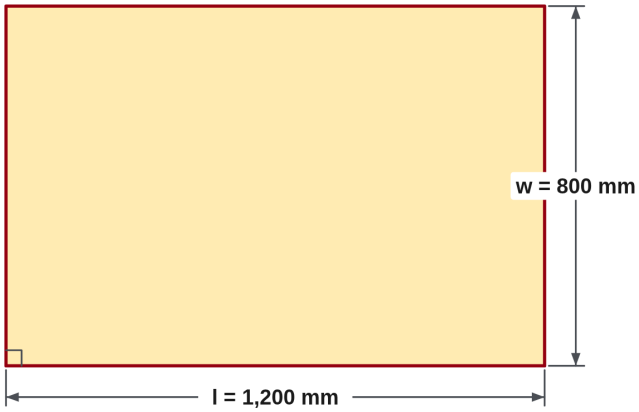
Choice (A–D)

Working / reason

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

06 CALCULATE

Read the Shape Canvas rectangle labels. The longer label is 1,200 mm and the shorter is 800 mm. Find their difference before applying any other operation.



Shape Canvas

Answer (mm)

Working / reason

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

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07 Calculate $(9 - 4)^2$.

Answer | **Working / reason**

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

08 Calculate $\sqrt{(49 + 32)}$.

Answer | **Working / reason**

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

09 Calculate $(6 + 2)^2$.

Answer | **Working / reason**

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

10 Calculate $100 - \sqrt{144}$.

Answer | **Working / reason**

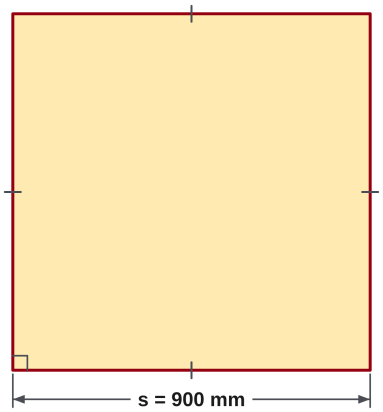
Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

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11 APPLY & CHECK

The Shape Canvas square has a 900 mm side label. Find the positive square root of the number 900 (no length formula is required).



Shape Canvas

Answer

Working / reason

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

12 A learner writes $(8 - 3)^2 = 8 - 9$. Which correction is right?

A. Subtract first, then square: $5^2 = 25$

B. Square only 8: $64 - 3 = 61$

C. Double the difference: $2 \times 5 = 10$

D. Add before squaring: $11^2 = 121$

Choice (A–D)

Working / reason

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

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13 Calculate √(16 + 9) + 7.

Answer	Working / reason

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

14 Calculate $2^2 \times (11 - 3)$.

Answer	Working / reason

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53

15 Calculate $(18 \div 3)^2 - 10$.

Answer	Working / reason

Textbook Chapter 3 pp.69–75; Chapter 2 pp.51–53